

THE NON-CENTRAL MULTIVARIATE BETA TYPE 2 DISTRIBUTION

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1. Introduction

$A(p \times p)$ is distributed according to a Wishart distribution with m degrees of freedom and $B(p \times p)$ is distributed independently according to a non-central Wishart with n degrees of freedom and non-centrality parameter Ω of rank $r \leq p$. We shall use the notation $A(p \times p) \sim W(\Sigma_1, m)$ and $B(p \times p) \sim W(\Sigma_2, n, \Omega)$.

The density function of B is given by (Constantine (1963))

$$(1.1) \quad f(B) = W(B; \Sigma_2, n) \exp(-\frac{1}{2} \text{tr } \Omega) \cdot {}_0F_1(\frac{1}{2}n; \frac{1}{4} \Omega \Sigma_2^{-1}), B > 0$$

where (Anderson (1958))

$$W(B; \Sigma_2, n) = [\Gamma_p(\frac{1}{2}n) |2\Sigma_2|^{\frac{1}{2}n-1} |B|^{\frac{1}{2}(n-p-1)} \cdot \exp(-\frac{1}{2} \text{tr } \Sigma_2^{-1}B).$$

See de Waal (1968) for further explanations.

In this paper the distribution of $V = B^{-1/2}AB^{-1/2}$, where the square root is symmetric, i.e. $B = (B^{1/2})^2$, will be derived. It will be shown that the densities of V and $|V|$ can be written as the product of independent beta densities of the second kind if $\Sigma_1 = \Sigma_2 = I$ and Ω is diagonal of rank $r < p$. In paragraph 3 the distribution of $\text{tr } V$ is derived in the case $\Sigma_1 = \Sigma_2 = \Sigma$ and Ω of rank $r < p$.

2. The distribution of V

The distribution of V can easily be derived from the joint distribution of A and B . The jacobian being,

$$J(A, B \rightarrow V, B) = |B|^{\frac{1}{2}(p+1)}.$$

Integration w.r.t. B in the joint density of V and B gives

$$(2.1) \quad f(V) = [\Gamma_p(\frac{1}{2}m) \Gamma_p(\frac{1}{2}n) |2\Sigma_1|^{\frac{1}{2}m} |2\Sigma_2|^{\frac{1}{2}n-1} \cdot \exp(-\frac{1}{2} \text{tr } \Omega) |V|^{\frac{1}{2}(m-p-1)} \int_{B>0} |B|^{\frac{1}{2}(m+n-p-1)} \cdot \exp[-\frac{1}{2} \text{tr } (\Sigma_2^{-1}B + B^{\frac{1}{2}} \Sigma_1^{-1}B^{\frac{1}{2}}V)] \cdot {}_0F_1(\frac{1}{2}n; \frac{1}{4} \Omega \Sigma_2^{-1}B) dB, V > 0.$$

Although this density function cannot be written in an explicit form, it will be used to derive some further results.

From 2.1 it follows that

$$(2.2) \quad E|V|^h = \Gamma_p(\frac{1}{2}m+h) |2\Sigma_1|^h [\Gamma_p(\frac{1}{2}m) \Gamma_p(\frac{1}{2}n) \cdot |2\Sigma_2|^{\frac{1}{2}n-1} \exp(-\frac{1}{2} \text{tr } \Omega) \int_{B>0} |B|^{\frac{1}{2}(n-2h-p-1)} \cdot \exp(-\frac{1}{2} \text{tr } \Sigma_2^{-1}B) {}_0F_1(\frac{1}{2}n; \frac{1}{4} \Omega \Sigma_2^{-1}B) dB,$$

recalling that

$$\int W(V; B^{\frac{1}{2}} \Sigma_1^{-1} B^{\frac{1}{2}}, m+2h) dV = 1.$$

Using the integral (Constantine (1963))

$$(2.3) \quad \int_{B>0} \exp(-\text{tr } RB) |B|^{a-1/2(p+1)} C_K(TB) dB \\ = \Gamma_p(a) |R|^{-a} (a)_K C_K(R^{-1}T), \text{ R and T symmetric we have}$$

$$(2.4) \quad E|V|^h = \Gamma_p(1/2 m+h) \Gamma_p(1/2 n-h) [\Gamma_p(1/2 m) \\ \cdot \Gamma_p(1/2 n)]^{-1} |\Sigma_1 \Sigma_2^{-1}|^h \exp(-1/2 \text{tr } \Omega) \\ \cdot {}_1F_1(1/2 n-h; 1/2 n; 1/2 \Omega)$$

If $\Sigma_1 = \Sigma_2 = 1$, the density function of V can be found explicitly from 2.1 by using 2.3, i.e.

$$(2.5) \quad f(V) = \Gamma_p(1/2(m+n)) [\Gamma_p(1/2 m) \Gamma_p(1/2 n)]^{-1} \\ \cdot \exp(-1/2 \text{tr } \Omega) |V|^{1/2(m-p-1)} |I+V|^{-1/2(m+n)} \\ \cdot {}_1F_1(1/2(m+n), 1/2 n; 1/2 \Omega (I+V)^{-1}), \quad V > 0.$$

According to the work of Troskie (1966), the distribution 2.5 will be called a non-central multivariate beta distribution of the second kind of full rank and denoted by

$$V(p \times p) \sim \beta_2(1/2 m, 1/2 n, 1/2 \Omega).$$

Hence, from 2.4, if $V \sim \beta_2(1/2 m, 1/2 n, 1/2 \Omega)$, then

$$(2.6) \quad E|V|^h = \Gamma_p(1/2 m+h) \Gamma_p(1/2 n-h) [\Gamma_p(1/2 m) \Gamma_p(1/2 n)]^{-1} \\ \cdot \exp(-1/2 \text{tr } \Omega) {}_1F_1(1/2 n-h, 1/2 n; 1/2 \Omega).$$

The h -th moment of $|I+V|^{-1}$ can be found from 2.5 by using the fact that $\int \beta_2(V; 1/2 m, 1/2 n, 1/2 \Omega) dV = 1$, i.e.

$$(2.7) \quad \int_{V>0} |V|^{a-1/2(p+1)} |I+V|^{-(a+b)} C_K(1/2 \Omega (I+V)^{-1}) dV \\ = \Gamma_p(a) \Gamma_p(b) [\Gamma_p(a+b)]^{-1} ((b)_K / (a+b)_K) C_K(1/2 \Omega).$$

Hence,

$$(2.8) \quad E|I+V|^{-h} = \Gamma_p(1/2 n+h) \Gamma_p(1/2(m+n)) [\Gamma_p(1/2 n) \\ \cdot \Gamma_p(1/2(m+n)+h)]^{-1} \exp(-1/2 \text{tr } \Omega) \\ \cdot {}_2F_2(1/2 n+h, 1/2(m+n)+h, 1/2 n; 1/2 \Omega).$$

An asymptotic distribution for $|I+V|^{-1}$, which is a function of Wilks criterion (Anderson (1958)), can be found in de Waal (1968).

If Ω is of rank $r < p$, i.e. there exists an orthogonal matrix C such that $C\Omega C' = \begin{pmatrix} \Omega_r & 0 \\ 0 & 0 \end{pmatrix}$, Ω_r of order r and diagonal, it seems difficult to write 2.5 in an explicit form. The reason is that 2.5 is not invariant under the orthogonal transformation $V \rightarrow CVC'$. The moments $|V|^h$ and $|I+V|^{-h}$ are however invariant and hence from 2.6 and 2.8 respectively

$$(2.9) \quad E|V|^h = \Gamma_p(\frac{1}{2}m+h)\Gamma_p(\frac{1}{2}n-h) [\Gamma_p(\frac{1}{2}m)\Gamma_p(\frac{1}{2}n)]^{-1} \\ \exp(-\frac{1}{2} \text{tr} \Omega_r) {}_1F_1(\frac{1}{2}n-h; \frac{1}{2}n; \frac{1}{2}\Omega_r)$$

and

$$(2.10) \quad E|I+V|^{-h} = \Gamma_p(\frac{1}{2}n+h)\Gamma_p(\frac{1}{2}(m+n)) [\Gamma_p(\frac{1}{2}(m+n)+h)\Gamma_p(\frac{1}{2}n)]^{-1} \\ \exp(-\frac{1}{2} \text{tr} \Omega_r) {}_2F_2(\frac{1}{2}n+h, \frac{1}{2}(m+n); \frac{1}{2}(m+n)+h, \frac{1}{2}n; \frac{1}{2}\Omega_r)$$

The reason why the hypergeometric function can be written as above, is that (James (1961))

$$C_K(\frac{1}{2}\Omega) = \begin{cases} 0 & K_{i>r} \neq 0 \\ C_K(\frac{1}{2}\Omega_r) & K_{i>r} = 0. \end{cases}$$

If we consider the densities of A and B in their canonical form however, i.e. $\Sigma_1 = \Sigma_2 = I$ and Ω diagonal, the density function 2.5 can be written as

$$(2.11) \quad f(V) = \Gamma_p(\frac{1}{2}(m+n)) [\Gamma_p(\frac{1}{2}m)\Gamma_p(\frac{1}{2}n)]^{-1} \exp(-\frac{1}{2} \text{tr} \Omega_r) \\ \cdot |V|^{\frac{1}{2}(m-p-1)} |I+V|^{-\frac{1}{2}(m+n)} \\ \cdot {}_1F_1(\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\Omega_r (I+V_{1,2})^{-1}), V > 0,$$

if Ω_r is the rank r , i.e. Ω_r diagonal of order r .

$$V_{1,2} = V_{11} - V_{12}(I+V_{22})^{-1}V_{21}$$

if V is partitioned as

$$(2.12) \quad V = \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{pmatrix}$$

such that V_{11} is of order r .

The distribution 2.11 will be called a non-central multivariate beta distribution of the second kind of rank r and denoted by

$$V(p \times p) \sim \beta_2(\frac{1}{2}m, \frac{1}{2}n, \frac{1}{2}\Omega_r).$$

It will now be shown that if $V \sim \beta_2(\frac{1}{2}m, \frac{1}{2}n, \frac{1}{2}\Omega_r)$, then the density function of V can be written as the product of independent beta distributions of the second kind.

Ignoring the constant term, 2.11 can be written

$$(2.13) \quad f(V) \propto \exp(-\frac{1}{2} \text{tr} \Omega_r) |V|^{\frac{1}{2}(m-p-1)} |I+V|^{-\frac{1}{2}(m+n)} \\ \cdot {}_1F_1((\frac{1}{2}m+n); \frac{1}{2}n; \frac{1}{2}\Omega_r (I+V_{1,2})^{-1}).$$

According to the partition 2.12, we have

$$|V| = |V_{22}| |V_{1,2} - V_{22}(I+V_{22})^{-1}V_{21}|.$$

$$\text{Let } U = V_{12}(I+V_{22})^{-1}V_{22}^{-1/2},$$

$$J(V_{12} \rightarrow U) = |(I+V_{22})^{-1}V_{22}^{-1}|^{-\frac{1}{2}r} \\ = |I+V_{22}|^{\frac{1}{2}r} |V_{22}|^{\frac{1}{2}r},$$

then

$$|V| = |V_{22}| |V_{1,2} - UU'|.$$

Let

$$T = V_{1,2}^{-1/2} U, J(U \rightarrow T) = |V_{1,2}|^{1/2} (p-r),$$

then

$$(2.14) \quad |V| = |V_{22}| |V_{1,2}| |I - TT'| : \\ J(V_{1,2} \rightarrow T) = J(V_{1,2} \rightarrow U) \cdot J(U \rightarrow T) \\ = |I + V_{22}|^{1/2r} |V_{22}|^{1/2r} |V_{1,2}|^{1/2} (p-r).$$

It also follows that

$$(2.15) \quad |I + V| = |I + V_{22}| |I + V_{1,2}|, J(V_{1,1} \rightarrow V_{1,2}) = 1.$$

Therefore, 2.13 becomes

$$(2.16) \quad f(V) a \exp(-1/2 \operatorname{tr} \Omega_r) |V_{1,2}|^{1/2} (m-r-1) |I + V_{1,2}|^{1/2} (m+n) \\ \cdot {}_1F_1(1/2(m+n); 1/2n; 1/2r (I + V_{1,2})^{-1}) \\ \cdot |V_{22}|^{1/2} (m-(p-r)-1) |I + V_{22}|^{-1/2} (m+n-r) \\ \cdot |I - TT'|^{1/2} (m-p-1),$$

i.e.

$$(2.17) \quad f(V) = f_1(V_{1,2}) \cdot f_2(V_{22}) \cdot f_3(T)$$

where

$$V_{1,2}(r \times r) \sim \beta_2(1/2m, 1/2n, 1/2\Omega_r),$$

$$V_{22}(p-r \times p-r) \sim \beta_2(1/2m, 1/2(n-r)),$$

(central beta of the second kind) and

$$f_3(T) a |I - TT'|^{1/2} (m-p-1).$$

From Anderson (1958), page 317, it follows that TT' will be distributed according to a central distribution of the first kind with $p-r$ and $m-p+r$ degrees of freedom if $r < p-r$. Hence, letting

$$V_r = (I - TT')^{-1/2} TT' (I - TT')^{-1/2}$$

it follows that

$$V_r(r \times r) \sim \beta_2(1/2(p-r), 1/2(m-p+r)).$$

Therefore it can be stated that

$$(2.18) \quad \beta_2(V(p \times p); 1/2m, 1/2n, 1/2\Omega_r) \\ = \beta_2(V_{1,2}(r \times r); 1/2m, 1/2n, 1/2\Omega_r) \\ \cdot \beta_2(V_{22}(p-r \times p-r), 1/2m, 1/2(n-r)) \\ \cdot \beta_2(V_r(r \times r); 1/2(p-r), 1/2(m-p+r)).$$

From 2.18 the densities of V_{22} and V_r can easily be written as the product of one-dimensional densities of the second kind by taking $\Omega = 0$ and $r = 1$. Because

of the large number of distributions, this result seems useless and will be omitted. In de Waal the joint density of V_{22} and T is written as the product of one-dimensional beta densities of the second kind in a simple form, but then the variables are not written in terms of the elements of V .

It can also be shown that $|V|$, if $V(p \times p) \sim \beta_2(\frac{1}{2}m, \frac{1}{2}n, \frac{1}{2}\Omega_r)$, is distributed as the product of independent beta variables of the second kind.

From 2.9 we have that

$$\begin{aligned} (2.19) \quad E|V|^h &= \Gamma_r(\frac{1}{2}m+h) \Gamma_r(\frac{1}{2}n-h) (\Gamma_r(\frac{1}{2}m) \Gamma_r(\frac{1}{2}n))^{-1} \\ &\quad \cdot \exp(-\frac{1}{2} \text{tr } \Omega_r) {}_1F_1(\frac{1}{2}n-h; \frac{1}{2}n; \frac{1}{2}\Omega_r) \\ &\quad \cdot \prod_{i=r+1}^p \frac{\Gamma(\frac{1}{2}(m+1-i)+h) \Gamma(\frac{1}{2}(n+1-i)-h)}{\Gamma(\frac{1}{2}(m+1-i)) \Gamma(\frac{1}{2}(n+1-i))} \\ &= E|G|^h \cdot \prod_{i=r+1}^p E(U_i)^h \end{aligned}$$

where

$$G(r \times r) \sim \beta_2(\frac{1}{2}m, \frac{1}{2}n, \frac{1}{2}\Omega_r)$$

and

$$u_i(1 \times 1) \sim \beta_2(\frac{1}{2}(m+1-i), \frac{1}{2}(n+1-i))$$

independent of each other. Hence, we can say that $|V|$ is distributed according to $|G| \cdot \prod_{i=r+1}^p u_i$. If $r = 1$, i.e. the linear case, then this result reduces to the result derived by Troskie (1966).

3. The distribution of $\text{tr } V$

Much attention has been given to the distribution of $\text{tr } V^{-1}$ in the central and non-central cases. The reason is that $\text{tr } V^{-1}$ is an important test criterium in multivariate regression and variance analysis (see for example Khatri and Pillai (1968)). Nothing is in fact known about the properties of $\text{tr } V$ which may also be considered as a test criterium, especially when the roots of V^{-1} are large. An exact distribution for $\text{tr } V$ will be derived of $\Sigma_1 = \Sigma_2 = \Sigma$ and Ω of rank $r \leq p$, but unfortunately the distribution exists only for $|\text{tr } V| < 1$.

The density function of $\text{tr } V$ will be defined by inverting the Laplace transform

$$g(t) = E \exp(-\text{tr } V)$$

of its density function. From 2.1

$$\begin{aligned} (3.1) \quad g(t) &= [\Gamma_p(\frac{1}{2}m) \Gamma_p(\frac{1}{2}n) |2\Sigma_1|^{\frac{1}{2}m} |2\Sigma_2|^{\frac{1}{2}n}]^{-1} \\ &\quad \cdot \exp(-\frac{1}{2} \text{tr } \Omega) \int_{B>0} \int_{V>0} |B|^{\frac{1}{2}(m+n-p-1)} \\ &\quad \cdot \exp(-\frac{1}{2} \text{tr } \Sigma_2^{-1}B) |V|^{\frac{1}{2}(m-p-1)} \exp[-\text{tr}(\frac{1}{2} B^{\frac{1}{2}} \Sigma_1^{-1} B^{\frac{1}{2}} + tI)V] \\ &\quad \cdot \sum_{k=0}^{\infty} \sum_K (C_K(\Omega \Sigma_2^{-1} B/4)/(\frac{1}{2}n) K_k!) \cdot dV dB. \end{aligned}$$

The integral w.r.t. V is

$$(3.2) \quad \int_{V>0} |V|^{\frac{1}{2}(m-p-1)} \exp[-\text{tr}(\frac{1}{2} B^{\frac{1}{2}} \Sigma^{-1} B^{\frac{1}{2}} + tI)V] dV$$

$$\begin{aligned}
&= \Gamma_p(1/2 m) |B|^{1/2} \Sigma_1^{-1} B^{1/2} + t|I|^{-1/2} m \\
&= \Gamma_p(1/2 m) \sum_{j=0}^{\infty} \Sigma_J t^{(1/2 mp+j)} (-1)^j (1/2 m)_J C_J(1/2 \Sigma_1^{-1} B) / j!
\end{aligned}$$

Substituting this in 3.1 for the integral w.r.t. V , we have

$$\begin{aligned}
(3.3) \quad g(t) &= (1/\Gamma_p(1/2 n) \exp(-1/2 \operatorname{tr} \Omega) |2 \Sigma_1|^{-1/2} m |2 \Sigma_2|^{-1/2} n \\
&\quad \cdot \int_{B > 0} |B|^{1/2 (m+n-p-1)} \exp(-1/2 \operatorname{tr} \Sigma_2^{-1} B) \sum_{k=0}^{\infty} \Sigma_K \sum_{j=0}^{\infty} \Sigma_J \\
&\quad \cdot (-1)^j t^{(1/2 mp+j)} (1/2 m)_J C_K(1/4 \Omega \Sigma_2^{-1} B) C_J(1/2 \Sigma_1^{-1} B) \\
&\quad \cdot [(1/2 n)_K k! j!]^{-1} dB.
\end{aligned}$$

Consider the integral w.r.t. B , i.e. say

$$\begin{aligned}
(3.4) \quad I &= \int_{B > 0} |B|^{1/2 (m+n-p-1)} \exp(-1/2 \operatorname{tr} \Sigma_2^{-1} B) \\
&\quad C_K(1/4 \Omega \Sigma_2^{-1} B) C_J(1/2 \Sigma_1^{-1} B) dB \\
&= |2 \Sigma_2|^{1/2 (m+n)} \int_{B > 0} |B|^{1/2 (m+n-p-1)} \exp(-\operatorname{tr} B) \cdot C_K(RB) C_J(TB) dB.
\end{aligned}$$

where $R = 1/2 \Sigma_2 - 1/2 \Omega \Sigma_2^{-1/2}$ (symmetric)

and $T = \Sigma_2^{1/2} \Sigma_1^{-1} \Sigma_2^{1/2}$ (symmetric)

This integral is unknown in the literature and a solution could not be found. Since R and T being symmetric matrices, let $B \rightarrow H B H'$, H an element of the orthogonal group with $\int_{O(p)} dH = 1$.

Hence

$$\begin{aligned}
(3.5) \quad I &= |2 \Sigma_2|^{1/2 (m+n)} \int_{B > 0} |B|^{1/2 (m+n-p-1)} \exp(-\operatorname{tr} B) \\
&\quad \cdot \int_{O(p)} C_K(R H B H) C_J(T H B H) dH dB
\end{aligned}$$

If we let $T = I_p$, i.e. $\Sigma_1 = \Sigma_2 = \Sigma$, integration over the orthogonal group can be performed, since (James (1964))

$$\int_{O(p)} C_K(R H B H) dH = C_K(R) C_K(B) / C_K(I_p)$$

Using the expansion (Constantine (1966))

$$C_K(B) C_J(B) = \sum_{\delta} g_{KJ}^{\delta} C_{\delta}(B)$$

3.5 becomes

$$\begin{aligned}
(3.6) \quad I &= |2 \Sigma|^{1/2 (m+n)} \sum_{\delta} g_{KJ}^{\delta} (C_K(R) / C_K(I_p)) \\
&\quad \int_{B > 0} |B|^{1/2 (m+n-p-1)} \exp(-\operatorname{tr} B) C_{\delta}(B) dB \\
&= \Gamma_p(1/2 (m+n)) |2 \Sigma|^{1/2 (m+n)} \sum_{\delta} g_{KJ}^{\delta} (1/2 (m+n))_{\delta} \\
&\quad C_K(R) C_{\delta}(I_p) / C_K(I_p)
\end{aligned}$$

Hence, in the case $\Sigma_1 = \Sigma_2 = \Sigma$, 3.3 becomes

$$(3.7) \quad g(t) = \frac{\Gamma_p(\frac{1}{2}(m+n))}{\Gamma_p(\frac{1}{2}n)} \exp(-\frac{1}{2} \text{tr } \Omega) \sum_{k=0}^{\infty} \sum_K \sum_{j=0}^{\infty} \sum_J \\ \Sigma_{\delta} \delta_{KJ} (-1)^j t^{(\frac{1}{2}mp+j)} (\frac{1}{2}m)_J (\frac{1}{2}(m+n))_{\delta} C_K(\frac{1}{2}\Omega) \\ C_{\delta}(I_p)/((\frac{1}{2}n)_K C_K(I_p) k! j!)$$

Using the Laplace inverse, i.e.

$$f(\text{tr } V) = (1/(2\pi i)) \int_{c-i\infty}^{c+i\infty} e^{-t(\text{tr } V)} g(t) dt$$

and integrating term-by-term, we have the following integral:

$$(3.8) \quad (1/(2\pi i)) \int_{c-i\infty}^{c+i\infty} e^{-t(\text{tr } V)} t^{(\frac{1}{2}mp+j)} dt \\ = (\text{tr } V)^{\frac{1}{2}mp+j-1} / \Gamma(\frac{1}{2}mp+j)$$

It can therefore be stated that if $A \sim W(\Sigma, m)$ and $B \sim W(\Sigma, n, \Omega)$ then the density function of $\text{tr } V$ is given by

$$(3.9) \quad f(\text{tr } V) = \frac{\Gamma_p(\frac{1}{2}(m+n))}{\Gamma_p(\frac{1}{2}n) \Gamma(\frac{1}{2}mp)} \exp(-\frac{1}{2} \text{tr } \Omega) \sum_{k=0}^{\infty} \sum_K \sum_{j=0}^{\infty} \\ \cdot C_K(\frac{1}{2}\Omega) C_{\delta}(I_p)/((\frac{1}{2}mp)_j (\frac{1}{2}n)_K C_K(I_p) k! j!)$$

The series being convergent for $|\text{tr } V| < 1$.

If Ω is of rank $r < p$, the distribution 3.9 stays the same except that Ω is replaced by Ω_r and K now consists of not more than r parts.

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